Kernix 0.7.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

14 workloads

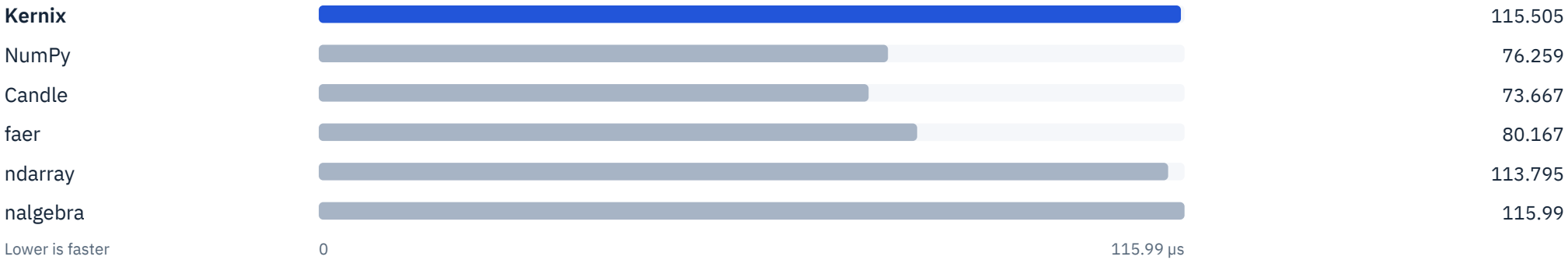
50 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

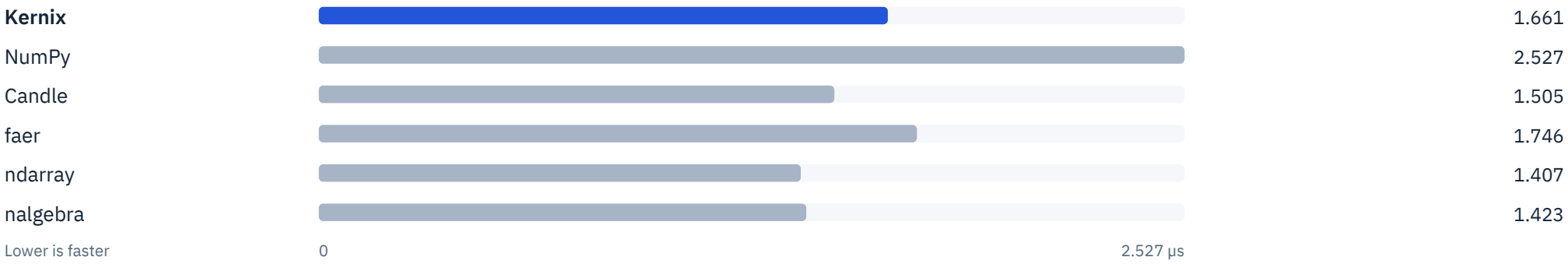
Kernix 0.7.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

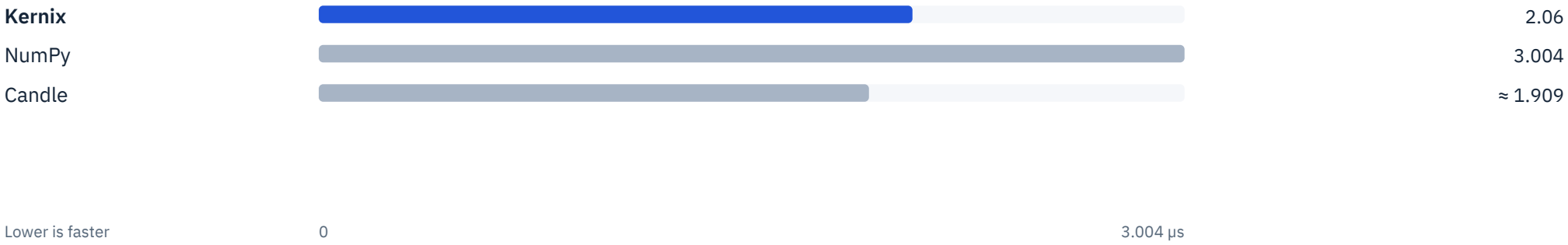
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

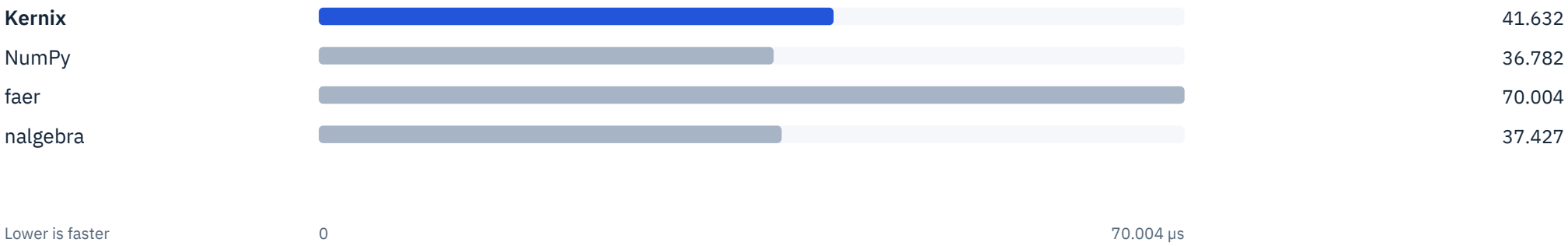


Kernix 0.7.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

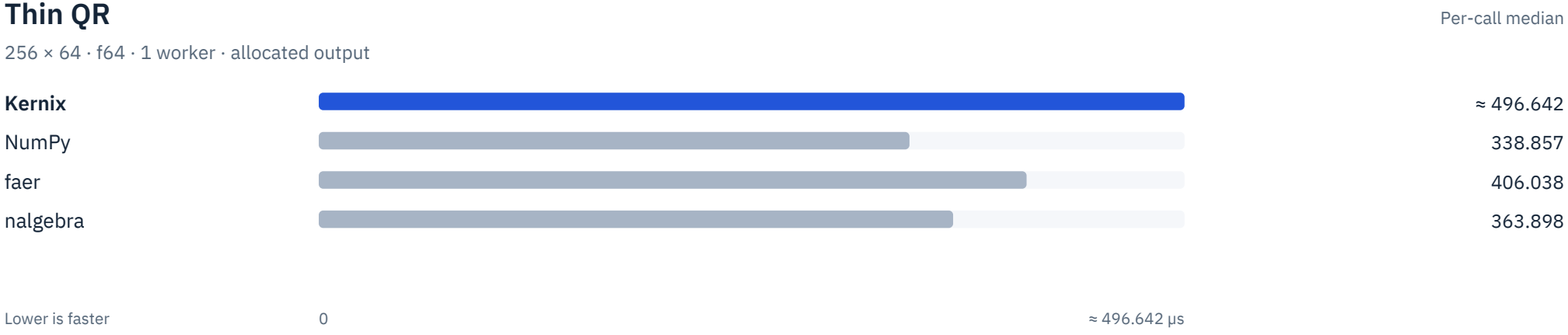
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output

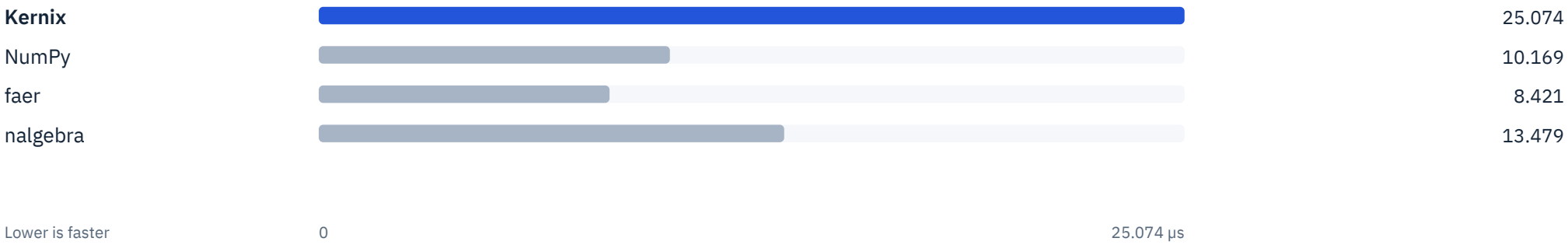


Kernix 0.7.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.7.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.7.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.7.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

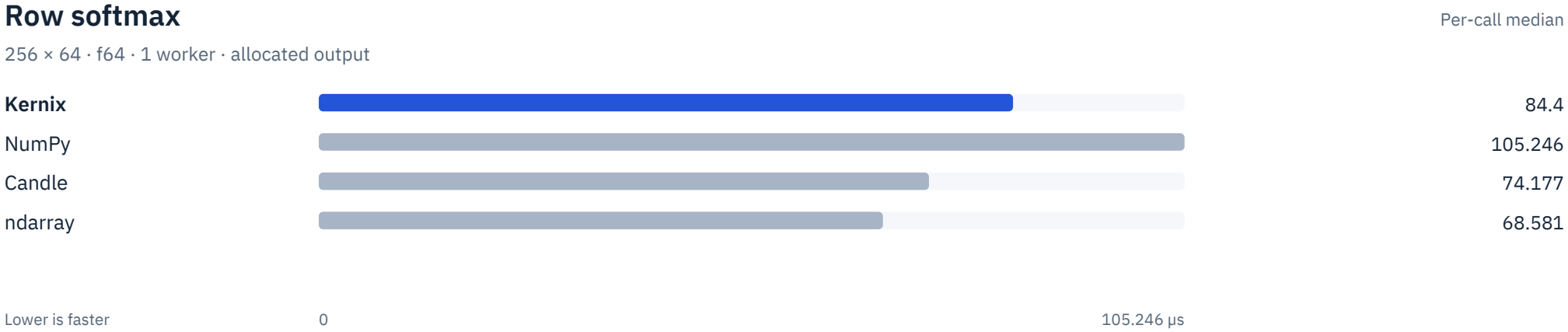


Kernix 0.7.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

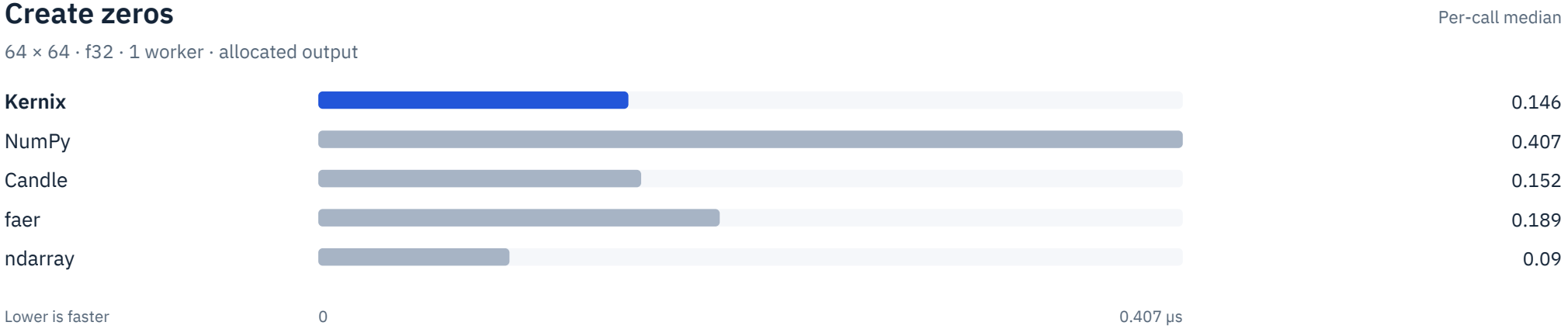


Kernix 0.7.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1661	1.661
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1407	1.407
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1423	1.423
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1746	1.746
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1505	1.505
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2527	2.527
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	115505	115.505
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	113795	113.795
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	115990	115.99
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	80167	80.167

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	73667	73.667
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	76259	76.259
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2060	2.06
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	1908.9999999999998	1.9089999999999998
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3004	3.004
Thin QR / Kernix 96 × 32 · f64	41632	41.632
Thin QR / nalgebra 96 × 32 · f64	37427	37.427
Thin QR / faer 96 × 32 · f64	70004	70.004
Thin QR / NumPy 96 × 32 · f64	36782	36.782
Thin QR / Kernix 256 × 64 · f64	496642.00000000006	496.64200000000006

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 256 × 64 · f64	363898	363.898
Thin QR / faer 256 × 64 · f64	406038	406.038
Thin QR / NumPy 256 × 64 · f64	338857	338.857
Cholesky factorization / Kernix 64 × 64 · f64	25074	25.074
Cholesky factorization / nalgebra 64 × 64 · f64	13479	13.479
Cholesky factorization / faer 64 × 64 · f64	8421	8.421
Cholesky factorization / NumPy 64 × 64 · f64	10169	10.169
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2403	2.403
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	3927	3.927
Model inference / Kernix batched normalized/sliced MLP · f64	3809	3.809

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	3282	3.282
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4646	4.646
Model recording + forward / Candle batched normalized/sliced MLP · f64	4756	4.756
Model backward / Kernix prepared graph → fresh gradients · f64	8668	8.668
Model backward / Candle prepared graph → fresh gradients · f64	13309	13.309
Parameter update / Kernix transactional SGD, fixed gradients · f64	482.99999999999994	0.48299999999999994
Row softmax / Kernix 256 × 64 · f64	84400	84.4
Row softmax / ndarray 256 × 64 · f64	68581	68.581
Row softmax / Candle 256 × 64 · f64	74177	74.177
Row softmax / NumPy 256 × 64 · f64	105246	105.246

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	146	0.146
Create zeros / ndarray 64 × 64 · f32	90	0.09
Create zeros / faer 64 × 64 · f32	189	0.189
Create zeros / Candle 64 × 64 · f32	152	0.152
Create zeros / NumPy 64 × 64 · f32	407	0.407
Create ones / Kernix 64 × 64 · f32	194	0.194
Create ones / ndarray 64 × 64 · f32	153	0.153
Create ones / faer 64 × 64 · f32	178	0.178
Create ones / Candle 64 × 64 · f32	216	0.216
Create ones / NumPy 64 × 64 · f32	1223	1.223